Physikalisches Anfängerpraktikum der Universität Heidelberg

Versuch 12

Moment of Intertia

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1 Introduction

1.1 Goal

The aim of this experiment is to investigate the rotational motion of a rotary pendulum. In particular, the directing torque D is determined using two independent methods. Furthermore, the moment of inertia of an irregularly shaped body is determined about different parallel axes, and the experimental results are compared with Steiner's theorem.

1.2 Physical Foundation

Rotational motion is described by equations analogous to those of linear motion. For a harmonic spring pendulum, the restoring force is proportional to the displacement,

$$F = -kx \tag{1}$$

which leads to the oscillation period

$$T = 2\pi\sqrt{\frac{m}{k}}. \tag{2}$$

For a rotary pendulum, the analogous quantity to the displacement is the angular displacement φ, and the restoring force is replaced by a directing torque. For small angular displacements, we assume the directing torque to be

$$M = -D\varphi, \tag{3}$$

where D is the directing torque of the spring system. The equation of motion of a rotary pendulum with moment of inertia J is consequently

$$J\varphi'' + D\varphi = 0, \tag{4}$$

which describes harmonic oscillations with the period

$$T = 2\pi\sqrt{\frac{J}{D}}. \tag{5}$$

This relation allows the directing torque to be determined from the oscillation period when the moment of inertia is known.

The moment of inertia depends on both the mass distribution of a body and the chosen axis of rotation. For a homogeneous disc with mass m_s and radius r_s, it is

$$J_s = \frac{1}{2}m_s r_s^2. \tag{6}$$

For an irregularly shaped body, the moment of inertia can be determined experimentally from its oscillation period. To investigate the dependence on the position of the rotation axis, Steiner's theorem is used. It states that the moment of inertia about an axis parallel to one through the centre of mass and displaced by a distance a is

$$J = J_u + Ma^2, \tag{7}$$

where J_u is the moment of inertia about the axis through the centre of mass and M is the mass of the body.

2 Messprotokoll

directing torque

Mass	angle	uncertainty
10g	0°	1°
60g	60°	1°
110g	110°	1°
160g	179°	3°
210g	239°	3°
260g	308°	3

Period Time Table Alone (20 oscilations) [S]

24,45	24,96	24,40

Period Time Table + disc (20 oscilations) [S]

33,40	33,42	33,39

Disc: Diameter: 10,57 ± 0,05 mm
Mass : 542 ± 1 g

Period Time Table + Shape (Offset = 0 mm, 20 oscilations) [S]

46,43	46,34	46,36

Shape: Mass: 665 ± 1 g

Period Time Table + Shape (offset = 5 mm, 20 oscilations) [S]

46,42	46,42	46,43

Period Time Table + Shape (offset = 10 mm, 20 oscilations) [S]

46,56	46,58	46,58

Period Time Table + Shape (offset = 20 mm, 20 oscilations) [S]

47,73	47,72	47,78

Period Time Table + Shape (offset = 30 mm, 20 oscilations) [S]

49,75	49,75	49,69

Period Time Table + Shape (offset = 40 mm, 20 oscilations) [S]

52,54	52,34	52,36

31.08.2026
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Abbildung 1: Protocol of the experiment

2.1 Sketch of the Experiment

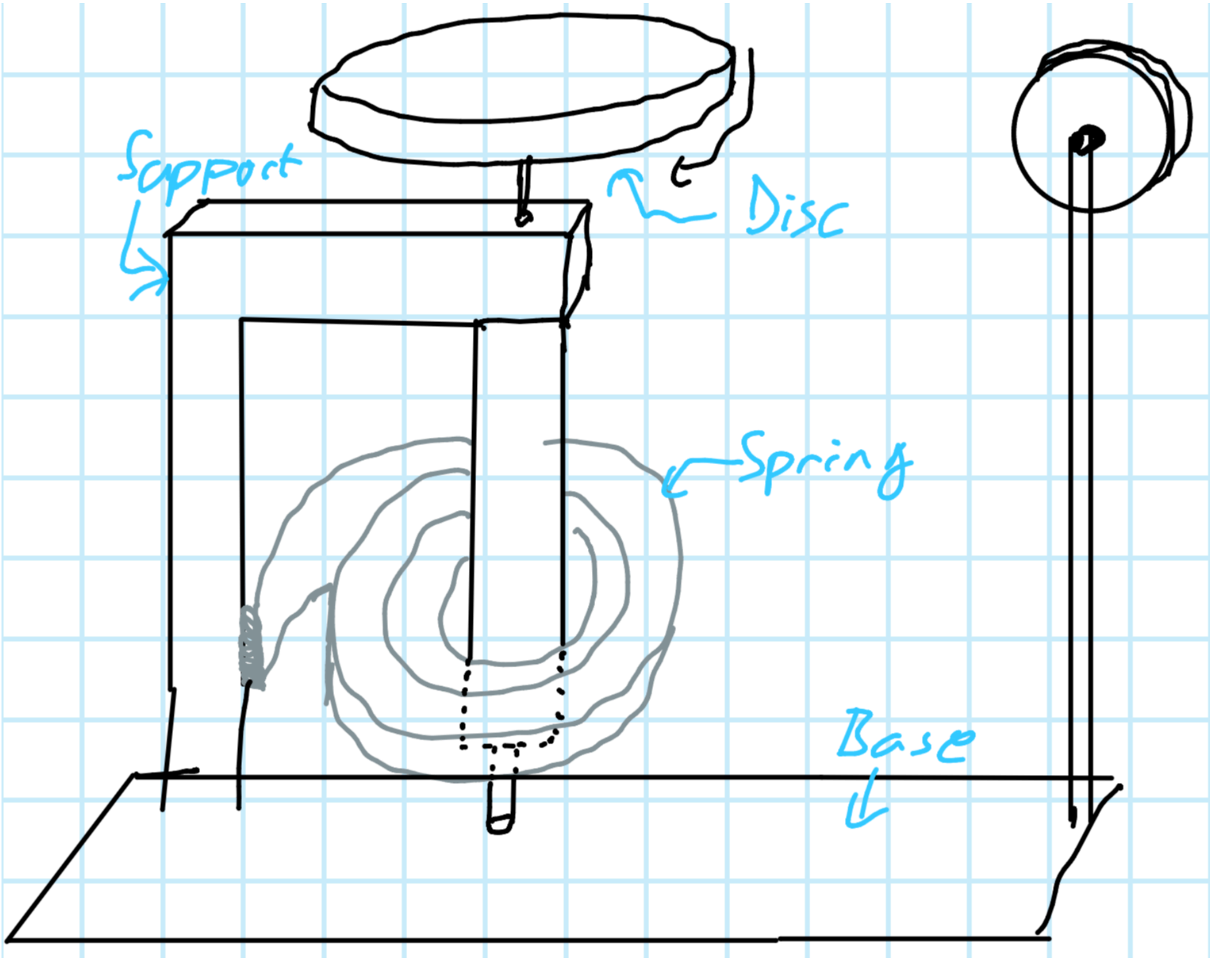

Abbildung 2: Sketch of the experiment

3 Auswertung

3.1 Determination of the Directing Torque

3.1.1 Method 1

To determine the directing torque statically, we measured the angle with verious weights pulling on a disk via a pully.

First, we need to calculate the torque that each weight exerts on the Pendulum using the following formula (The radius of the disk is 5cm):

$$M = m * g * 5 \text{ cm} \tag{8}$$

We found out online that the scale has an error of $\pm 1g$ and we assumed the error during the experiment to be at least 1°, with some measurements having a bigger error due to static friction at the equilibrium. There, we could not be sure where the true equilibrium was, because the pendulum would not move in a range of a few degrees. We noted this in our protocoll when this was the case.

We chose to ignore the error that the scale contributes to the Torque measurement, because it is a lot smaller than the error of the angle measurement.

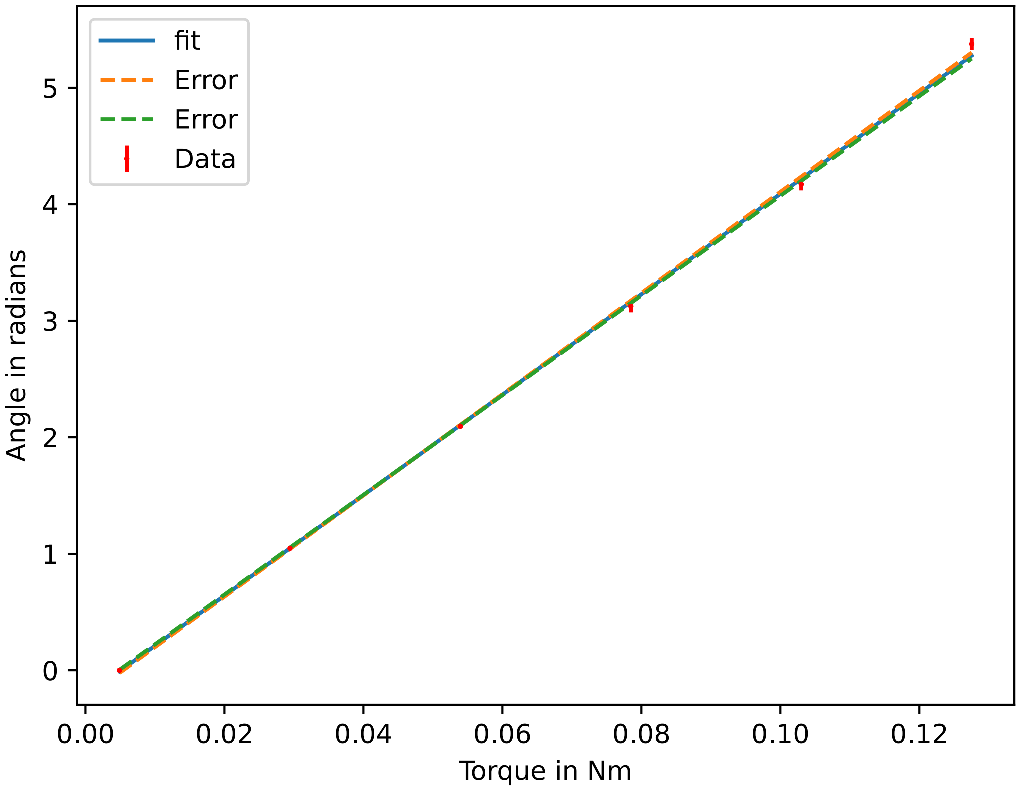

Abbildung 3: Plot of the Angle to

Slope: $a_1 = (43.10 \pm 0.32)\frac{\mathrm{rad}}{\mathrm{Nm}}$ (Determined using `scipy`)

From this we can calculate the Directing Torque:

$$\boxed{D_1 = \frac{1}{a_1} = (0.023\,20 \pm 0.000\,17)\,\mathrm{N\,m\,rad^{-1}} \qquad (9)}$$

3.1.2 Method 2

For our second measurement of the Directing Torque, we measured the Period of the Pendulum with and without a disk with known Moment of Inertia.

The disk has a mass of $(542 \pm 1)\,\mathrm{g}$ and a radius of $(525.5 \pm 0.25)\,\mathrm{mm}$. The uncertainties are from the scale and from the caliper.

Using the Formula for the Moment of Inertia of a Disk, we get

$$J_S = \frac{1}{2} m_s r_s^2 = (0.000\,748 \pm 0.000\,007)\,\mathrm{kg\,m^2} \tag{10}$$

We used python package `uncertainties`, which calculates the Error analytically, by hand we have to calculate the total derivative of J_s:

$$\mathrm{d}J_s = \frac{1}{2} r_s^2\,\mathrm{d}m + m_s r_s\,\mathrm{d}r_s \tag{11}$$

Using the Laws of error propagation, we get:

$$\Delta J_s = \sqrt{\left(\frac{1}{2} r_s^2 \Delta m\right)^2 + (m_s r_s \Delta r_s)^2} \approx 7.3 \cdot 10^{-6} \text{ kg m}^2 \tag{12}$$

We measured the time of 20 periods using a stopwatch (device error negligble). We estimated the error due to our reaction time to be about 200 ms.

We added this (according to the rules of error propagation) to the average error of the mean value $S_M = \frac{S_E}{\sqrt{N}}$ (determined from the 3 measurements):

$$\Delta T = \sqrt{(200\,\mathrm{ms})^2 + S_M^2} \tag{13}$$

For our measurements, we got a Period of

$$\text{Without Disk: } T_1 = (1.222 \pm 0.010)\,\mathrm{s} \tag{14}$$

$$\text{With Disk: } T_2 = (1.670 \pm 0.010)\,\mathrm{s} \tag{15}$$

As discussed in the Introduction, we have the following Formulas for T_1 and T_2 (assuming J_T to be the moment of inertia of the table):

$$T_1 = 2\pi \sqrt{\frac{J_T}{D}} \tag{16}$$

$$T_2 = 2\pi \sqrt{\frac{J_S + J_T}{D}} \tag{17}$$

We can solve both equations for J_T:

$$J_T = D \frac{T_1^2}{4\pi^2} \tag{18}$$

$$J_T = D \frac{T_2^2}{4\pi^2} - J_s \tag{19}$$

This gives us:

$$D \frac{T_1^2}{4\pi^2} = D \frac{T_2^2}{4\pi^2} - J_s \tag{20}$$

$$\frac{4\pi^2 J_s}{D} = T_2^2 - T_1^2 \tag{21}$$

$$D = \frac{4\pi^2 J_s}{T_2^2 - T_1^2} \tag{22}$$

After substituting the values, we get:

$$\boxed{D_2 = (0.0228 \pm 0.0008)\,\mathrm{N\,m\,rad^{-1}}} \tag{23}$$

3.1.3 Error Propagation by Hand

$$\mathrm{d}D = \frac{4\pi^2}{T_2^2 - T_1^2}\,\mathrm{d}J_s + (4\pi^2 J_s)\left(\frac{2T_1\,\mathrm{d}T_1}{(T_2^2 - T_1^2)^2} - \frac{2T_2\,\mathrm{d}T_2}{(T_2^2 - T_1^2)^2}\right) \tag{24}$$

$$\Delta D = \sqrt{\left(\frac{4\pi^2}{T_2^2 - T_1^2}\Delta J_s\right)^2 + (4\pi^2 J_s)^2\left(\left(\frac{2T_1\Delta T_1}{(T_2^2 - T_1^2)^2}\right)^2 + \left(\frac{2T_2\Delta T_2}{(T_2^2 - T_1^2)^2}\right)^2\right)} \tag{25}$$

(We actually verified it by hand that the result matches with the one from python, it was not fun...)

3.2 Moment of Ineratia of Irregular Shape

To determine the Moment of Intertia of the Irregular Shape J_u, we did the same procedure as above to determine the Period along with its error:

$$T_u = (2.319 \pm 0.010)\,\mathrm{s} \tag{26}$$

From rearranging the Formula above, we get

$$\boxed{J_u = \frac{D}{4\pi^2} * (T_u^2 - T_1^2) = (0.002\,283 \pm 0.000\,035)\,\mathrm{kg\,m^2}} \tag{27}$$

The error propagation works analogously to the exercise above.

3.3 Steiners Theorem

After calculating the average value and incorporating the statistical and reaction time error, we measured the following Periods:

Distance a to the center of mass in mm	**Period** T_u in s
5	(2.32 ± 0.01)
10	(2.33 ± 0.01)
20	(2.39 ± 0.01)
30	(2.49 ± 0.01)
40	(2.64 ± 0.02)

We can now calculate the moments of inertia by rearranging Gleichung 22 to:

$$J_{i_e} = \frac{D}{4\pi^2} * (T_2^2 - T_1^2) \tag{28}$$

(With J_{i_e} being the experimentally determined moment of intertia.)

By using our best known value of D (obtained in Gleichung 9) and our known value of T_1, we can get the different moments of inertia by setting T_2 to the different values of T_u corresponding to the distances from the center of mass.

Additionally, we used Steiner's Theorem to calculate the moment of inertia J_{i_S} for the various distances a_i from the center of mass:

$$J_{i_S} = J_u + M \cdot a_i^2 \tag{29}$$

Distance a to the center of mass in mm	J_{i_e}	J_{i_S}
5.0	$(0.002\,288 \pm 0.000\,035)$	$(0.002\,299 \pm 0.000\,035)$
10.0	$(0.002\,309 \pm 0.000\,035)$	$(0.002\,349 \pm 0.000\,035)$
20.0	$(0.002\,47 \pm 0.000\,04)$	$(0.002\,549 \pm 0.000\,035)$

30.0	$(0.002\,75 \pm 0.000\,04)$	$(0.002\,881 \pm 0.000\,035)$
40.0	$(0.003\,21 \pm 0.000\,06)$	$(0.003\,347 \pm 0.000\,035)$

Plotting these values as a function of a^2 yields the following plot:

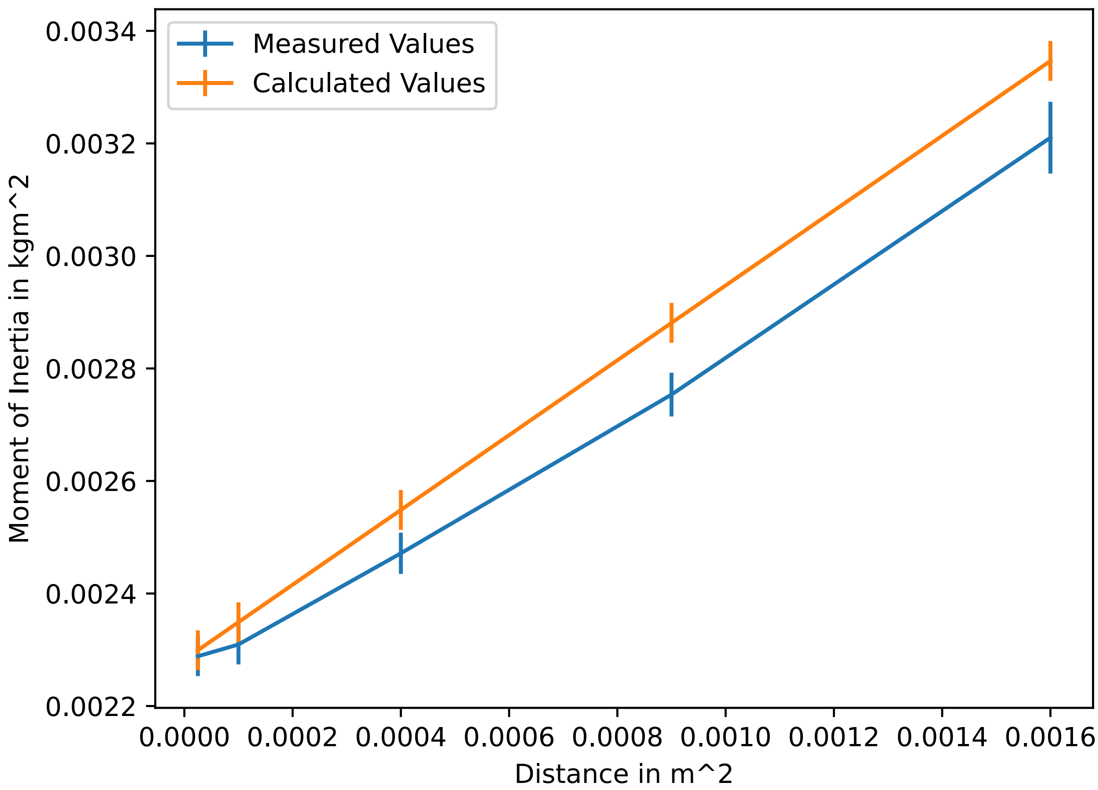

Abbildung 4: Moment of Inertia as a function of the distance to the center of mass squared

4 Results

$$\text{Directing Torque determined statically } D_1 = (0.023\,20 \pm 0.000\,17)\,\mathrm{N\,m\,rad^{-1}} \tag{30}$$

$$\text{Directing Torque determined with the oscillation } D_2 = (0.0228 \pm 0.0008)\,\mathrm{N\,m\,rad^{-1}} \tag{31}$$

$$\text{Moment of Intertia of the unknown Shape: } J_u = (0.002\,283 \pm 0.000\,035)\,\mathrm{kg\,m^2} \tag{32}$$

5 Discussion

5.1 Directing Torque

We determined the directing torque of the spring with two different methods. The standardized Difference is

$$\beta_D = \frac{D_1 - D_2}{\sqrt{\Delta D_1^2 + \Delta D_2^2}} \approx 0.49 \tag{33}$$

This indicates that the wo values are well inside the error range of each other.

Next, I am going to point out possible sources of Error that we did not account for:

5.1.1 Possible Sources of Error for D_1

- Friction in the pully and the bearing of the pendulum

5.1.2 Possible Sources of Error for D_2

- Systematic error in the time measurement

5.2 Moment of Intertia

To estimate if our value is plausible, I calculated the radius that a disk with the same moment of inertia would have using the following formula:

$$R_{\text{equiv}} = \sqrt{\frac{2J_u}{M}} \approx 9.2 \text{ cm} \tag{34}$$

This seems plausible, because the irregular shape had some parts with smaller radius, but also some with bigger radius.

5.3 Steiners Theorem

We can observe in Abbildung 4, that the experimentially determined values deviate from the expected values.

A likely cause could be the inaccuracy of the measurement of the center of mass. To investigate this, we plotted the values again after subtracting and adding a few milimeters to the Distance. In the following Plot we shifted the Distance towards the center of mass by 3mm:

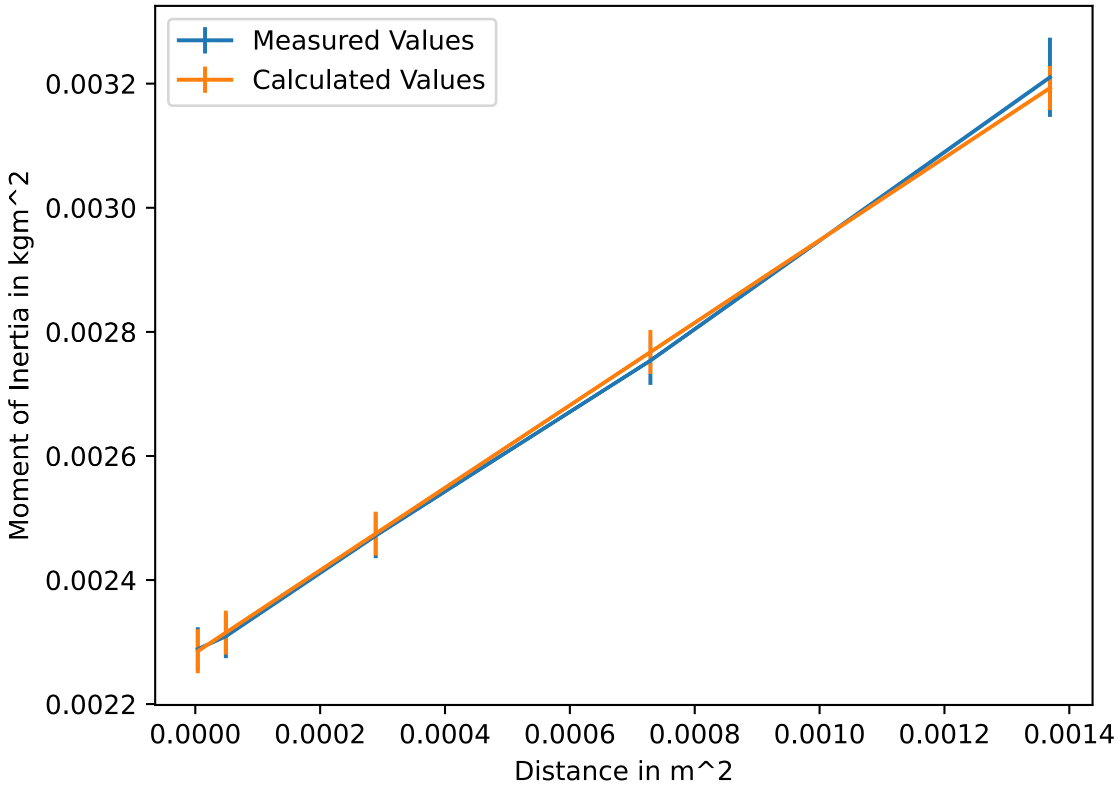

Abbildung 5: Moment of Inertia as a function of the distance to the center of mass squared (with the distance shifted towards the center of mass by 3mm)

We can see that the experimental data fits the calculation a lot better here. Possible explanations for this error could be:

- inaccurate marking of the center of mass
- inaccurate positioning of the shape on the pendulum (unlikely, because not systematic)
- inaccurate marking of the distances from the center of mass (unlikely, because not systematic)